SOFTWARE ENGINEERING ACADEMIC PROJECT — MID-SEMESTER EVALUATION

Department of Computer Science & Engineering

Individual Contribution Record & Declaration

Project: AutoTestHub (Web Automation & Testing Platform)

TASK ALLOCATION • GIT AUDIT TRAIL • ACADEMIC INTEGRITY DECLARATION

PROJECT METADATA

System Name: AutoTestHub

Repository Branch: `feature/execution-service`

Total Commits (Mid-Sem): 14 Structured Commits

Active Contributors: 4 Group Members

DEVELOPMENT TEAM

Shristi Neekhra [Reg No: XXXXXXXX]

Shifa Siddiqui [Reg No: XXXXXXXX]

Shruti Gupta [Reg No: XXXXXXXX]

Ishree [Reg No: XXXXXXXX]

Supervisor: [Supervisor Name]

1. Individual Contribution Breakdown

All team members have contributed equitably across the software development lifecycle, with clearly demarcated subsystem ownership and end-to-end integration responsibilities.

Member Name	Assigned Subsystem	Specific Technical Responsibilities	Approx. LOC	Contribution %
Shristi Neekhra	Integration Pipeline & Execution Service	• Implemented asynchronous `executionService.js` subprocess bridge. • Managed stdin JSON piping and stdout regex parsing. • Built standalone integration pipeline tester (`testExecution.js`). • Coordinated full branch merges and mid-semester release packaging.	~450 LOC	**25%**
Shifa Siddiqui	Backend Gateway & Database Architecture	• Initialized Node.js Express server and MongoDB connection. • Designed Mongoose `TestCase` and `TestResult` schemas. • Developed REST APIs (`POST /api/tests` , `GET /api/tests` , `POST /api/tests/:id/run`). • Implemented controller error handling and database index configurations.	~400 LOC	**25%**
Shruti Gupta	Selenium Engine & Automation Worker	• Designed core Python worker architecture and runner CLI. • Integrated Selenium 4 Selenium Manager and Chrome lifecycle management. • Implemented dynamic action handlers (`OPEN_URL` , `TYPE` , `CLICK` , `ASSERT_TITLE`). • Built in-memory validation suite and local HTTP integration test harness.	~650 LOC	**25%**
Ishree	Frontend Dashboard & Control Desk	• Designed responsive Control Desk interface (HTML5 / CSS3). • Implemented real-time terminal output streaming visualizer. • Built client-side simulation playback mode. • Implemented speed controls, theme toggling, and log export utilities.	~450 LOC	**25%**

2. Git Commit History Audit Trail

The following verified commit log reflects active development across all project branches leading up to the mid-semester integration:

Commit Hash	Author	Date & Time	Commit Message & Summary
`a587b46`	Shristi Neekhra	2026-10-19 16:08	`integration pipeline` : Connected Express routes with executionService and Python worker.
`c7f4f59`	Shristi Neekhra	2026-10-19 15:54	Merge remote-tracking branch 'origin/frontend' into feature/execution-service.
`350c38d`	Shristi Neekhra	2026-10-19 15:54	Merge remote-tracking branch 'origin/frontend' into feature/execution-service.
`7b96d36`	Shristi Neekhra	2026-10-19 15:53	Merge remote-tracking branch 'origin/feature/selenium-engine' into feature/execution-service.
`ccab87f`	Ishree	2026-10-19 14:47	`Add gitignore` : Excluded frontend environment and runtime dependencies.
`d55fb75`	Ishree	2026-10-19 14:47	`Add project documentation` : Documented Control Desk setup and execution instructions.
`9eba1ff`	Ishree	2026-10-19 14:47	`Add project dependencies` : Specified Flask and Selenium requirements.
`1958384`	Ishree	2026-10-19 14:47	`Add Flask application` : Implemented Control Desk server and real Chrome runner.
`9604413`	Ishree	2026-10-19 14:46	`Add frontend JavaScript` : Built terminal streaming and task execution dispatcher.
`7df8ca1`	Ishree	2026-10-19 14:46	`Add frontend CSS` : Styled Control Desk layout, dark mode theme, and components.
`74213ed`	Ishree	2026-10-19 14:46	`Add frontend HTML` : Created semantic markup for Control Desk and viewport.
`cbc3da4`	Shifa Siddiqui	2026-10-19 13:22	`backend initialized, mongodb connected, add testcase and testresult model and apis` .
`42a31ab`	Shruti Gupta	2026-10-18 16:48	`feat(selenium): implement complete Selenium automation engine` with runner and tests.
`d7fb35e`	Shifa Siddiqui	2026-10-17 17:16	`Initial commit` : Created repository baseline.

3. Academic Integrity & Contribution Declaration

We, the undersigned members of the AutoTestHub project group, hereby confirm and declare that:

1. All submitted code, architectural designs, test cases, and project documentation are the original work of this group, developed in accordance with the university's academic integrity policies.
2. Each member has actively contributed to the tasks outlined in the contribution breakdown above, and each member is capable of fully demonstrating and explaining the implementation of their respective components.
3. No confidential credentials, API keys, or production passwords have been committed or bundled into the submitted materials.
4. Use of generative AI tools for boilerplate scaffolding, formatting, and debugging has been fully and transparently disclosed in Section 4 below, and all generated artifacts were rigorously reviewed, tested, and modified by the students.

Signatures & Confirmations

Member Name	Student Registration Number	Role & Contribution	Signature / Status
Shristi Neekhra	[Reg No: XXXXXXXX]	Integration Lead & Execution Service	*Confirmed • 2026-10-19*
Shifa Siddiqui	[Reg No: XXXXXXXX]	Backend & Database Lead	*Confirmed • 2026-10-19*
Shruti Gupta	[Reg No: XXXXXXXX]	Selenium Engine Architect	*Confirmed • 2026-10-19*
Ishree	[Reg No: XXXXXXXX]	Frontend Lead & Control Desk	*Confirmed • 2026-10-19*

4. Generative AI Tools Disclosure (Section 6 Compliance)

Tool Name	Purpose of Use	Nature of Generated Content	Verification / Modification by Students
Gemini / Antigravity	Documentation structuring, SVG diagram scaffolding, and unit test edge case brainstorming.	Markdown formatting, SVG vector path boilerplate, and sample JSON payload templates.	All generated code and test assertions were manually verified, executed against real Chrome instances, debugged, and integrated into the project codebase by group members.